

SWIR-HSI ANALYSIS

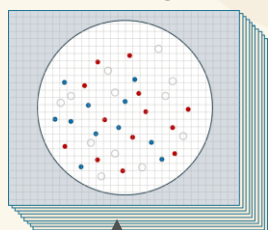
Cellulose



Silicon

Known concentrations of 30µm red PE, blue PE and 70µm white PVC were spiked into water and filtered using cellulose filters and silicon filters. Filters were imaged in SWIR-HSI camera (Specim, Finland, 1000-2500 nm).

HSI Image



n number of MPs

Set cut-off

1 pixel, 1 HSI spectrum

Photons detected at a pixel



Compare spectra

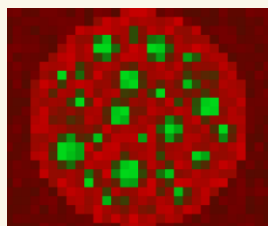
Target method for Anomaly Detection: SAM

Reference HSI spectrum

E.g., pure plastic, non-plastic, etc.



Similarity map computed against reference spectra



Summary of Decisions

Major decisions:

- ❖ Reference Spectra
- ❖ Preprocessing
- ❖ Similarity Threshold
- ❖ Particle size (# pixel)

All depends on:

Sample

Filter

Plastic type

Particle Size

Resolution

The trap

We tuned the pipeline using the answer. In real analysis, the answer is what we are trying to find.

Core Question Remains:

Are we counting the correct pixels?

Perspective

The next step for the microplastics community is not just more proof-of-detection studies, but the establishment of spectroscopic and chemometric frameworks to **validate** detection and enable reliable analysis.

How to validate? Your thoughts here:

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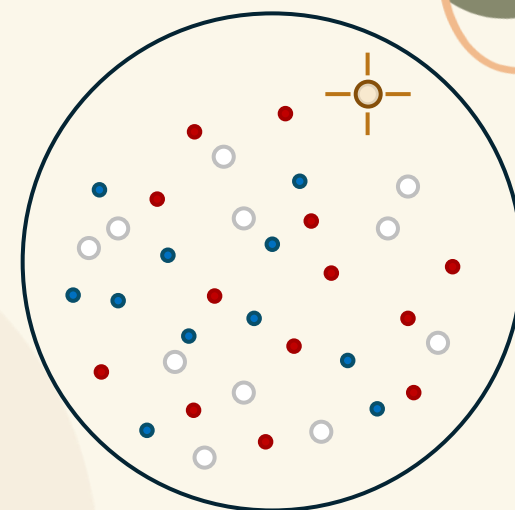


References:

- Amigo, J. M., Babamoradi, H., & Elcoroaristizabal, S. (2015). Hyperspectral image analysis. A tutorial. *Analytica chimica acta*, 896, 34-51.
- Rinnan, A., Van Den Berg, F., & Engelsen, S. B. (2009). Review of the most common pre-processing techniques for near-infrared spectra. *TrAC Trends in Analytical Chemistry*, 28(10), 1201-1222.

Towards Validated SWIR Hyperspectral Methods for Small Microplastic ($\leq 100 \mu\text{m}$) Detection in Water Filtration Samples: Challenges and Perspectives

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Are we counting the correct pixels?

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Challenge

One Pixel $\neq$ One Pure Spectrum

The signal at a single pixel depends on chemistry and on other factors such as particle geometry, surface texture, light penetration, scattering, surroundings, and noise (Amigo et al., 2015).

For small MPs, these effects are amplified. Thin particles transmit more background signal, pixels covering particles at or below pixel size are spectrally mixed, and edge scattering has a greater influence (Fig. 1, Fig. 2).

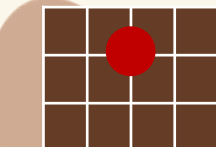
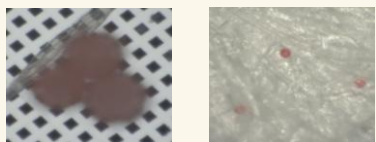


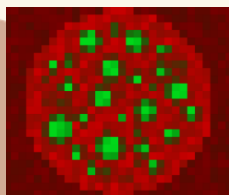
Fig. 1. Particle covering 4 pixels (top) and resulting mixed pixels (bottom).

Fig. 2. uRaman images of PE red on silicon (left) and cellulose (right) filter. (Some PE particles go under the cellulose fiber)

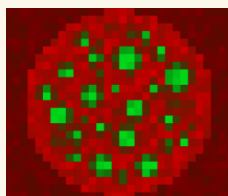
Perspective

Addressing Pixel-Mixing and Scattering

1. Row-wise preprocessing: reduces scatter effects
2. Include mixed spectra as a reference so pixels are compared against realistic plastic-on-background spectra



SAM vs pure plastic



SAM vs plastic on a filter

Fig. 3. Illustrative examples, not actual results

Challenge

No Single Preprocessing Recipe

There are many preprocessing options broadly falling into scatter-correction methods (e.g., SNV) and spectral derivatives (e.g., Savitzky-Golay derivatives) (Rinnan, et al., 2009). There is no single method that guarantees enhancement of spectral contrast between MPs and the imaging background, and the optimal choice depends on the data (Fig. 4).

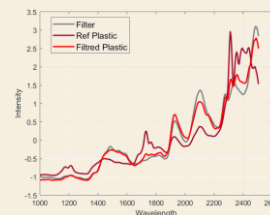
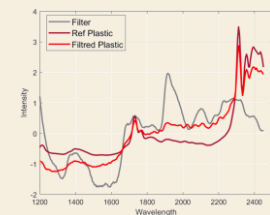


Fig. 4. Spectra of red PE against silicon (top) and cellulose (bottom) after SNV preprocessing.

Perspective

Tailor preprocessing based on needs



Case 1: Plastic and Filter are very different

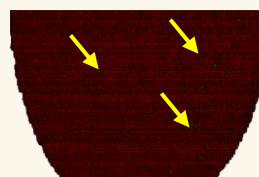
Fig. 5.a. PE on Silicon

SNV works

Case 2: Plastic and Filter are very similar

Fig. 5.b. PE on Cellulose

2D + SNV works



Case 3: Mixed plastics, hard to differentiate

Fig. 5.c. PE and PV on Cellulose Filter

2D + SNV works



Challenge

Pixel $\neq$ Particle: What is a particle?

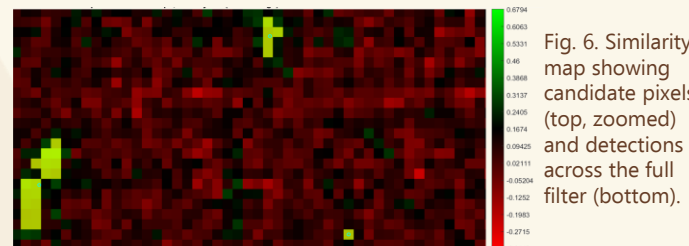


Fig. 6. Similarity map showing candidate pixels (top, zoomed) and detections across the full filter (bottom).

Even after accounting for pixel mixing and preprocessing, two questions remain: What minimum size defines a particle in an image (1, 2, 3 pixels? More?), and what spectral similarity threshold should be used for counting? (Fig. 6)

Perspective

On setting the cut-off and particle counting

1. Make an informed decision based on the pixel limit, particle size (Fig. 1 & 6), and the surroundings, e.g., filter influence (Fig. 2).
2. Report counts as a proxy for the number of particles only (Fig. 7)
3. Unknown: estimate, report limits

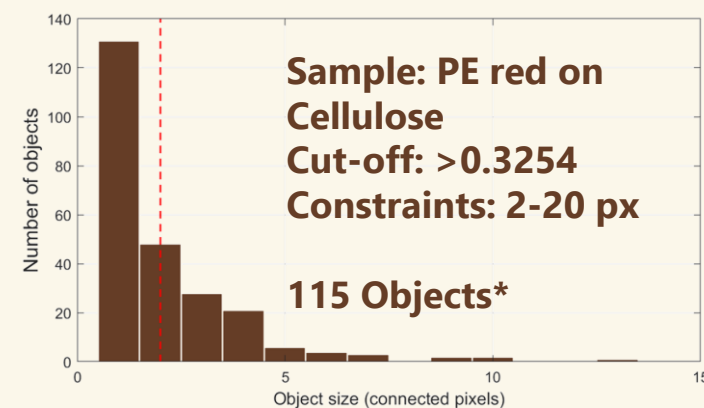


Fig. 7. Object size distribution; dashed line marks the 2-pixel minimum.